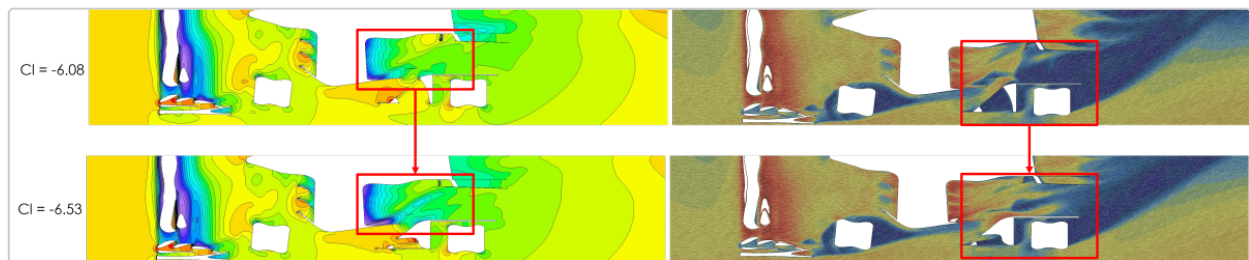


Cooler Housing

CH, incl. WP rear

Concept / Season Summary

Getting the cooler housing wrong can really hurt improvements. As shown in the picture the old design may have caused separation over the Wheelplate (WP) which increased pressure in that region which strongly influenced the pressure zone under the Mainwing. Removing the sharp corner from this region not only solved separation but also increased pressure difference over the cooler itself with a stronger stagnation point in front of the radiator.



Also the outlet is positioned higher on the EP, so that the dirty air does not influence the underside of the FP. Having the outlet too low did cause separation on corner inside which lost around 0.06 cl.

As a metric for comparison I used the projected inlet/outlet area on the x and y plane. Closing the inlet influences the mass flow more than closing the outlet. Smaller inlet and outlet both increase cl, but the outlet has a bigger impact because of the sensitive influence on the underside of the FP. Closing the intake gives you more air around the CH which strengthens the lower pressure zone around the curvature into the rear WP, which then supports the mainwing.

We aimed for a massflow around 0.1 kg/s, which we kinda decided on for ourselves because we didn't get a straight value from the cooling guy.

NOTE

Cooler simulations are highly dependent on simulation parameters. With no validation these results are probably more like best guesses.